

IMPLEMENTATION OF 3DES

AIM

To implement the **3DES (Triple Data Encryption Standard)** technique for encrypting and decrypting a given message using three keys.

ALGORITHM

1. Start the program.
2. Read the plaintext and three keys from the user.
3. Encrypt the plaintext using **Key 1**.
4. Encrypt the result using **Key 2**.
5. Encrypt the result again using **Key 3**.
6. Display the encrypted text.
7. Decrypt using **Key 3**, **Key 2**, and **Key 1** in reverse order.
8. Display the decrypted plaintext.
9. Stop the program.

PROGRAM

```
#include <stdio.h>
```

```
#include <string.h>
```

```
int main() {
```

```
    char text[100];
```

```
    char key1[9], key2[9], key3[9];
```

```
    char encrypted[100], decrypted[100];
```

```
    int i, len;
```

```
    printf("Enter Plain Text: ");
```

```
    fgets(text, sizeof(text), stdin);
```

```
    text[strcspn(text, "\n")] = '\0';
```

```
    printf("Enter Key 1 (8 characters): ");
```

```
scanf("%8s", key1);
```

```
printf("Enter Key 2 (8 characters): ");
```

```
scanf("%8s", key2);
```

```
printf("Enter Key 3 (8 characters): ");
```

```
scanf("%8s", key3);
```

```
len = strlen(text);
```

```
for (i = 0; i < len; i++) {
```

```
    encrypted[i] = text[i] ^ key1[i % 8];
```

```
    encrypted[i] = encrypted[i] ^ key2[i % 8];
```

```
    encrypted[i] = encrypted[i] ^ key3[i % 8];
```

```
}
```

```
printf("\nEncrypted Text: ");
```

```
for (i = 0; i < len; i++)
```

```
    printf("%02X", (unsigned char)encrypted[i]);
```

```
for (i = 0; i < len; i++) {
```

```
    decrypted[i] = encrypted[i] ^ key3[i % 8];
```

```
    decrypted[i] = decrypted[i] ^ key2[i % 8];
```

```
    decrypted[i] = decrypted[i] ^ key1[i % 8];
```

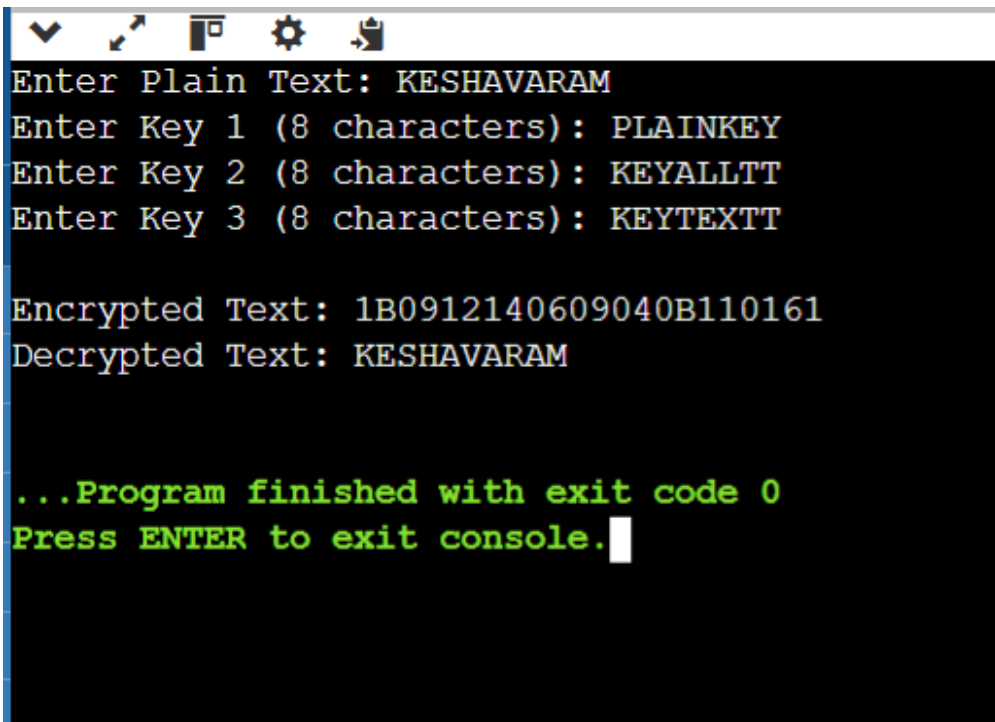
```
}
```

```
decrypted[len] = '\0';
```

```
printf("\nDecrypted Text: %s\n", decrypted);
```

```
    return 0;  
}
```

OUTPUT

A screenshot of a terminal window with a black background and white and green text. The window has a title bar with standard icons. The text shows the user entering 'KESHAVARAM' as plain text, followed by three 8-character keys: 'PLAINKEY', 'KEYALLTT', and 'KEYTEXTT'. The program then outputs the encrypted text '1B0912140609040B110161' and the decrypted text 'KESHAVARAM'. At the end, it displays a green message: '...Program finished with exit code 0' and 'Press ENTER to exit console.' with a cursor.

```
Enter Plain Text: KESHAVARAM  
Enter Key 1 (8 characters): PLAINKEY  
Enter Key 2 (8 characters): KEYALLTT  
Enter Key 3 (8 characters): KEYTEXTT  
  
Encrypted Text: 1B0912140609040B110161  
Decrypted Text: KESHAVARAM  
  
...Program finished with exit code 0  
Press ENTER to exit console.
```

RESULT

Thus, the **3DES encryption and decryption process** was implemented successfully, and the original plaintext was recovered after decryption.

